

1. In the derivation of the equation governing the bending of light due to a massive object we had the following equation

$$\frac{d^2 v}{d\phi^2} + v = \epsilon v^2$$

where $v = b/r$ and $\epsilon = 3M/b$ in geometric units. Using ϵ as the perturbation parameter, we expressed the solution as $v = v_0 + \epsilon v_1$ and substitute in the differential equation. Show that the equation in linear order in ϵ is

$$\frac{d^2 v_1}{d\phi^2} + v_1 = \epsilon v_0^2 = \epsilon \sin^2 \phi$$

Show that $v_1 = A + B \sin^2 \phi$ is an appropriate ansatz for the solution and determine A, B .

2. The Friedman-Robertson-Lemaitre-Walker metric is

$$ds^2 = c^2 dt^2 - a(t)^2 \left(\frac{d\chi^2}{1 - k\chi^2} + \chi^2 d\Omega^2 \right)$$

- (a) Compute the Riemann and Ricci tensors for this metric.
- (b) Using Einstein's equations, for $k = 0$, show that the Friedmann equations are

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3} T_0^0 \quad \frac{2\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} = 8\pi G T_1^1 \quad (1)$$

3. In Quiz 2, you saw that the time taken in one revolution in a circular orbit as measured in the Schwarzschild coordinate follows Kepler's law. Compute the corresponding proper time.
4. In class, we discussed about inclusion of λ term in Einstein's equations. The resultant of this term was to bring a force of repulsion at longer distances and thus providing static universe. To understand this slightly better, let us consider that the force between two bodies is given by

$$F \equiv \left(\frac{-GM}{r^2} + \lambda r c^2 \right) m$$

- (a) Obtain an equilibrium distance where the equilibrium condition is attained.
- (b) Is this stable equilibrium? Argue.

5. A distant galaxy has a redshift $z \equiv \delta\lambda/\lambda$ of 0.2. According to Hubble's law, how far away was the galaxy when the light was emitted if the Hubble constant is $72(km/sec)/Mpc$?
6. Suppose in another theory of gravity, the metric outside a spherical star is given by

$$ds^2 = \left(1 - \frac{2M}{r}\right) [dt^2 - dr^2 - r^2 d\Omega^2]$$

Calculate the deflection of light by a spherical star in this theory assuming that the photons move along the null geodesics.

7. In Newtonian mechanics, Kepler's area law says that equal areas are swept in equal times. Does it hold true by either the proper time or Schwarzschild time for the orbital radius $R > 2M$?
8. Obtain the null geodesics in 2+1 flat space-time obtained by the polar coordinates.

$$ds^2 = dt^2 - dr^2 - r^2 d\Omega^2$$